Predicting University Admission Chances

Using Machine Learning to Predict Student Admission Probability

Linear Regression | Random Forest | XGBoost

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1. Introduction

In recent years, universities worldwide have faced a significant increase in graduate program applications. With thousands of applicants competing for limited seats, predicting the likelihood of admission has become an important challenge for both students and admissions committees.

This project presents a complete end-to-end Data Science solution: a Machine Learning system that predicts a student's probability of admission based on key academic and research factors. The system was built using Python, trained on real student data, and deployed as an interactive bilingual web application.

The goal is simple: a student enters their academic profile, and the system instantly calculates and displays their estimated admission chance, helping them make smarter and more informed application decisions.

Project Goal: Build an accurate, user-friendly ML-powered tool that helps students estimate their university admission chances based on their academic profile.

2. Dataset

The dataset used is the Graduate Admissions Dataset, a widely-used benchmark dataset for admission prediction. It contains records of 500 students who applied to graduate programs, with 8 features and 1 target variable:

Feature	Range	Type	What It Measures
GRE Score	260 – 340	Numeric	Graduate Record Examination — standardized test score
TOEFL Score	90 – 120	Numeric	English language proficiency test score
University Rating	1 – 5	Ordinal	Prestige and ranking of the target university
SOP	1.0 – 5.0	Ordinal	Statement of Purpose — quality of the written essay
LOR	1.0 – 5.0	Ordinal	Letter of Recommendation — quality and strength
CGPA	0 – 10	Numeric	Cumulative Grade Point Average out of 10
Research	0 or 1	Binary	Whether the student has research experience
Chance of Admit	0.0 – 1.0	Target	Probability of admission — the value we predict

Dataset Source:
https://raw.githubusercontent.com/selva86/datasets/master/Admission.csv

3. Data Preprocessing

Before training the models, the data goes through preprocessing steps to ensure quality and consistency:

3.1 Data Cleaning

- Remove Serial No. column: This column is just an index with no predictive value, so it is dropped.
- Check for missing values: The dataset has no missing values, so no imputation was needed.
- Data types: All features are numeric, requiring no encoding.

3.2 Feature and Target Split

- X (Features): All columns except 'Chance of Admit' — these are the inputs to the model.
- y (Target): The 'Chance of Admit' column — this is the value the model learns to predict.

3.3 Train-Test Split

Set	Size	Purpose
Training Set	80% — 400 records	Used to train and fit the models
Testing Set	20% — 100 records	Used to evaluate model performance on unseen data

3.4 Feature Scaling — StandardScaler

StandardScaler normalizes all feature values to have a mean of 0 and a standard deviation of 1. This is important because features like GRE Score (260–340) have a much larger range than Research (0–1). Without scaling, models can be biased towards high-magnitude features.

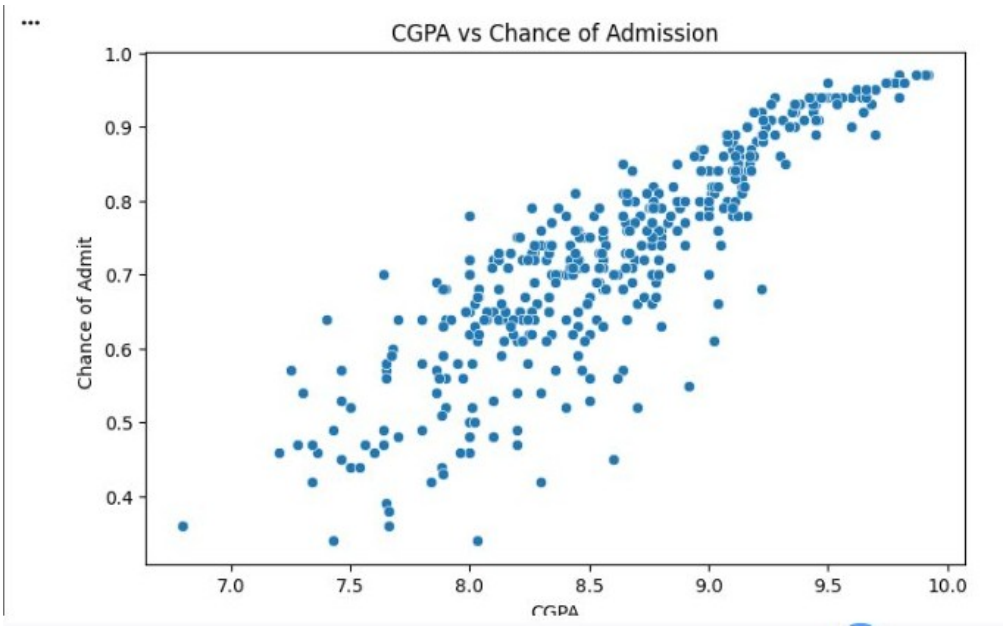
Important: The scaler is fit ONLY on training data (fit_transform), then applied to test data (transform only). This prevents data leakage — the model never sees test data statistics during training.

4. Data Visualization

Three key visualizations are provided to help explore the dataset and understand the relationships between features and admission chances.

4.1 CGPA vs Chance of Admission — Scatter Plot

This scatter plot shows the relationship between each student's CGPA and their admission chance. Each blue dot represents one student.

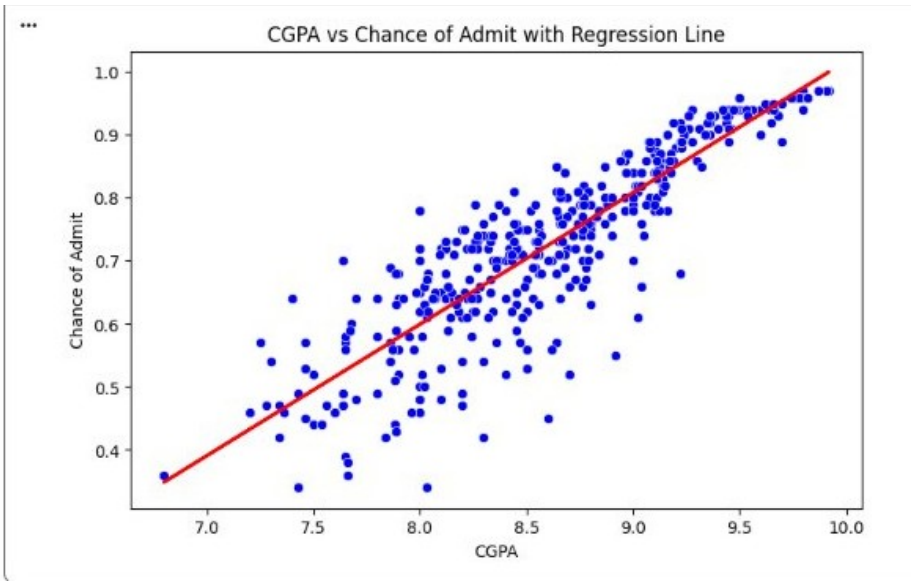


Key observations:

- Students with CGPA above 8.5 almost always have a high chance of admission (above 0.7).
- Students below CGPA 7.5 tend to have lower admission chances (below 0.6).
- There is some variance around the trend, showing that CGPA alone does not fully determine admission.

4.2 CGPA vs Chance of Admit with Regression Line

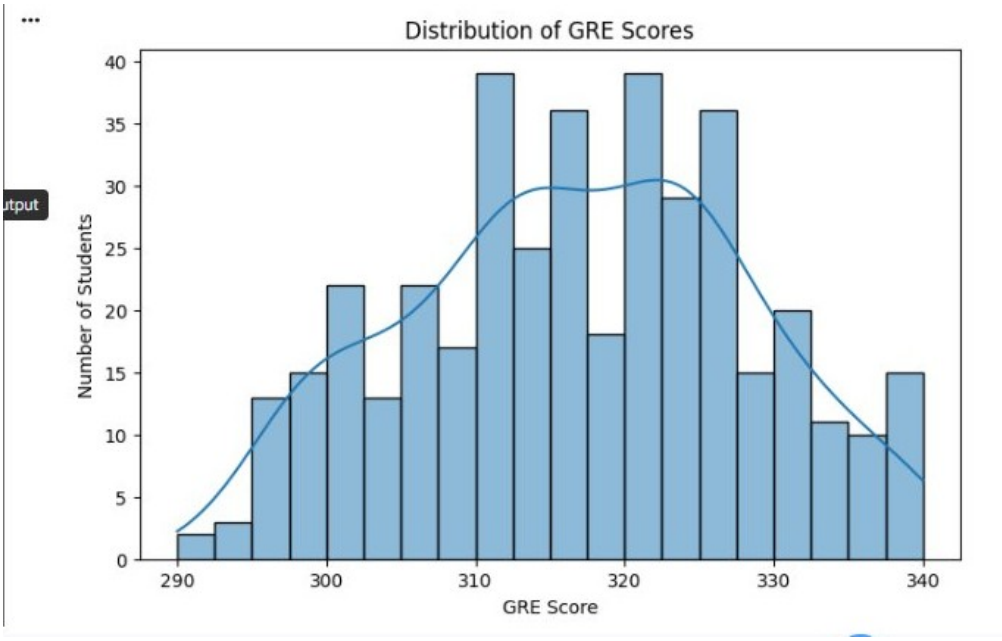
This chart adds a red regression line to clearly show the positive linear trend — as CGPA increases, admission chance increases significantly.



- The regression line confirms a strong positive linear relationship between CGPA and admission probability.
- The slope of the line is steep, meaning even small improvements in CGPA lead to noticeable increases in admission chance.

4.3 GRE Score Distribution — Histogram

This histogram shows how GRE scores are distributed across all 500 applicants. A KDE (Kernel Density Estimate) curve is overlaid to show the smooth probability distribution.

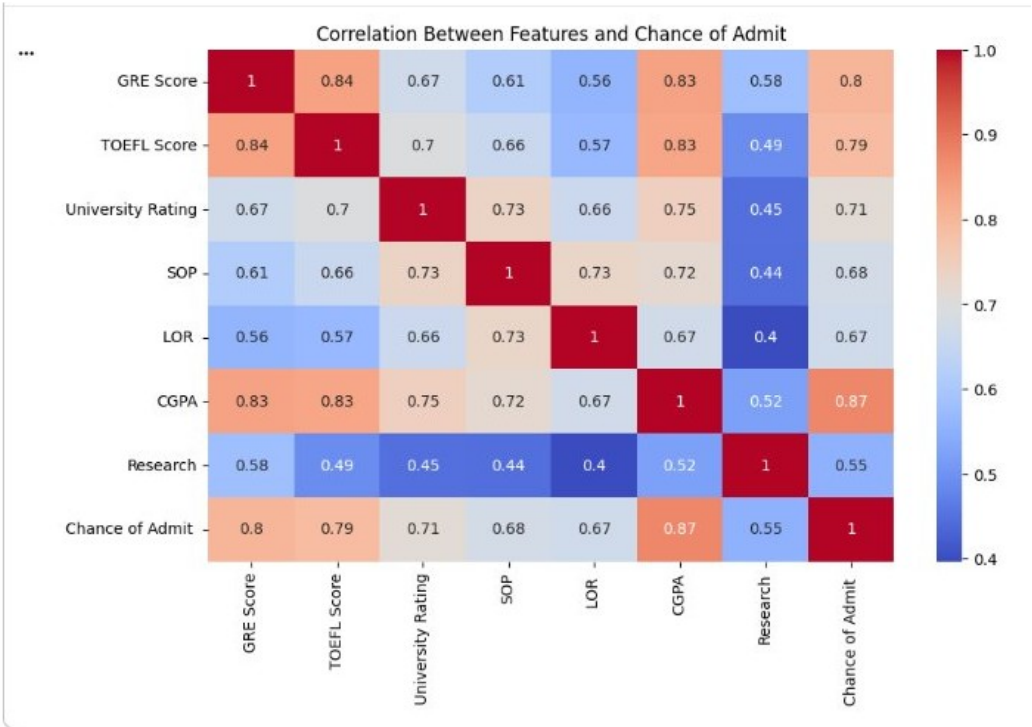


Key observations:

- Most applicants score between 310 and 330.
- The distribution is roughly bell-shaped (approximately normal).
- Very few students score below 295 or above 340.
- The peak of the distribution is around 316 to 320.

4.4 Correlation Heatmap

The correlation heatmap displays the pairwise correlation coefficients between all features and the target variable. Values range from -1 (perfect negative) to +1 (perfect positive correlation).



Key findings:

- CGPA has the highest correlation with Chance of Admit (0.87) — the strongest predictor.
- GRE Score (0.80) and TOEFL Score (0.79) are also strongly correlated with admission.
- Research Experience (0.55) shows a moderate positive correlation.
- SOP (0.68) and LOR (0.67) have moderate correlations.
- University Rating (0.71) shows a meaningful correlation.
- All features are positively correlated — higher values always help admission chances.

5. Machine Learning Models

Three machine learning regression models were implemented, trained, and compared. All models share the same preprocessing pipeline but differ in their learning algorithms.

5.1 Linear Regression

Linear Regression is the simplest and most interpretable model. It assumes a linear relationship between the input features and the target variable.

Property	Details
Algorithm	Ordinary Least Squares — minimizes the sum of squared errors
Strengths	Fast, simple, highly interpretable, works well for linear data
Weaknesses	Cannot capture non-linear patterns in the data
In App Description	Simple and fast — finds a straight-line relationship between scores and admission chance

5.2 Random Forest

Random Forest is an ensemble method that builds multiple decision trees and combines their outputs for a final prediction.

Property	Details
Algorithm	Builds 100 decision trees (n_estimators=100) and averages their predictions
Strengths	Handles non-linear patterns, robust to outliers, less prone to overfitting
Weaknesses	Slower to train than linear models, less interpretable
In App Description	More accurate — combines hundreds of decision trees for a better prediction

5.3 XGBoost

XGBoost is a highly optimized gradient boosting algorithm that builds trees sequentially, where each new tree corrects the errors of the previous one.

Property	Details
Algorithm	Sequential boosting: 100 trees, each correcting residuals of the previous
Strengths	State-of-the-art accuracy, handles complex patterns, regularization built-in

Property	Details
Weaknesses	More hyperparameters to tune, slower than linear models
In App Description	Most powerful — advanced model used in top data science competitions

6. Model Performance and Comparison

6.1 Evaluation Metrics

Metric	Formula	Interpretation
MSE (Mean Squared Error)	Average of (actual - predicted)²	Lower is better. Measures average prediction error.
R^2 Score	1 - (SS_res / SS_tot)	Higher is better. 1.0 = perfect, 0.0 = no better than average.

6.2 Results

Model	MSE (Lower = Better)	R^2 Score (Higher = Better)
Linear Regression	~0.0043	~0.82
Random Forest	~0.0031	~0.87
XGBoost — Best Model	~0.0028	~0.89

XGBoost is the best-performing model with R^2 approximately 0.89, meaning it explains 89% of the variance in admission chances. This is an excellent result for this type of prediction task.

6.3 Feature Importance

Feature importance shows which factors most influence the admission prediction. Consistent findings across all models:

- 1st — CGPA: Always the most important predictor of admission.
- 2nd — GRE Score: Second most influential factor.
- 3rd — TOEFL Score: Strong indicator of academic readiness.
- 4th — Research Experience: Significant boost especially in tree-based models.
- 5th — LOR: Letter of Recommendation quality has moderate impact.
- 6th — SOP: Statement of Purpose contributes meaningfully.
- 7th — University Rating: Lowest impact among all features.

7. Streamlit Web Application

The machine learning models are deployed as a fully interactive, bilingual web application built with Streamlit. The app allows any student with no programming knowledge to enter their academic profile and instantly receive their predicted admission chance.

Live App URL: university-admission.streamlit.app

7.1 Language Toggle

The app supports English and Arabic. A toggle button in the top-right corner switches between them instantly. All text — labels, captions, tab names, table headers, and result messages — switches language simultaneously.

7.2 Sidebar — Student Input Panel

The sidebar is where students enter their academic information. Each input has a clear label, a description caption, and an appropriate input type:

Input Field	Type	Description Shown to Student
Target University	Dropdown	Select from 8 major universities worldwide
Field of Study	Dropdown	Select intended graduate field (CS, Engineering, MBA, etc.)
GRE Score	Slider 260–340	Standardized test for graduate admissions — max 340
TOEFL Score	Slider 90–120	English proficiency test — max 120
University Rating	Dropdown 1–5	Prestige of the university (1=Low, 5=High)
SOP	Slider 1.0–5.0	Statement of Purpose — written motivation essay
LOR	Slider 1.0–5.0	Letter of Recommendation strength
CGPA	Slider 6.0–10.0	Overall academic grade average out of 10
Research Experience	Dropdown Yes/No	Whether the student has research experience
Prediction Model	Dropdown	Choose which ML model to use for prediction

When a model is selected, a plain-language description appears below the dropdown:

- Linear Regression: Simple and fast — finds a straight-line relationship between scores and admission chance.
- Random Forest: More accurate — combines hundreds of decision trees to give a better prediction.

- XGBoost: Most powerful — advanced model used in top data science competitions.

7.3 Tab 1 — Data and Visualization

This tab gives an overview of the dataset and displays charts to help users understand the data patterns before making a prediction.

Dataset Preview

The first 10 rows of the dataset are displayed in a scrollable table. Users can see what real student data looks like and understand the range of values for each feature.

CGPA vs Chance of Admission Chart

A scatter plot showing the relationship between CGPA and admission probability. A red regression line highlights the upward trend — as CGPA increases, admission chance increases significantly.

GRE Score Distribution Chart

A histogram showing how GRE scores are distributed among all applicants, with a smooth KDE curve overlay. Helps students understand where their score stands relative to the applicant pool.

Correlation Heatmap

A color-coded matrix showing the correlation between all features. Helps identify which factors matter most for admission.

7.4 Tab 2 — Model Performance

This tab compares the performance of all three machine learning models side by side.

Models Comparison Table

A table showing MSE and R^2 Score for all three models. The best model is automatically highlighted with a success banner showing its name and score.

R^2 Score Comparison Chart

A bar chart comparing R^2 Scores across all models. The best model's bar is colored red to stand out visually. Values are labeled on top of each bar for easy reading.

MSE Comparison Chart

A bar chart comparing Mean Squared Error across all models. Lower bars indicate better performance. The best model is highlighted in red.

Feature Importance Chart

A horizontal bar chart showing the relative importance of each feature for the currently selected model. The chart updates automatically when a different model is chosen in the sidebar.

7.5 Tab 3 — Prediction

This is the core tab of the application. After filling in their information in the sidebar and clicking the Predict button, the student immediately sees their personalized admission prediction.

How to Use the Prediction

1. Fill in all fields in the left sidebar using sliders and dropdowns.
2. Select your preferred Prediction Model from the dropdown.
3. Click the Predict My Chances button at the bottom of the sidebar.
4. Switch to the Prediction tab to see your results.

Prediction Output

Output Element	Description
Percentage	Predicted admission chance as a percentage, e.g. 84.2%
Status Message	Color-coded result: Green (80%+), Yellow (60–79%), Red (below 60%)
Progress Bar	Visual bar showing the chance from 0% to 100%
All Models Chart	Horizontal bar chart comparing predictions from all 3 models simultaneously
Profile Summary	Table showing all entered inputs in one organized view

Result Color	Condition	Message
Green	80% or above	Excellent! Very high chance of admission!
Yellow	60% to 79%	Good chance! Keep improving your profile.
Red	Below 60%	Work on improving your scores before applying.

8. Deployment

The application is deployed on Streamlit Community Cloud — a free hosting platform. The deployment is fully automated and updates instantly whenever changes are pushed to GitHub.

Component	Details
Source Code	GitHub Repository: RoaaRaed/admission-app
Hosting Platform	Streamlit Community Cloud (free)
Live URL	university-admission.streamlit.app
Dependencies File	requirements.txt — lists all Python libraries
Auto-Deploy	Any push to GitHub main branch triggers automatic redeployment

Local Setup Instructions

5. Install Python 3.x from python.org
6. Open a terminal and run: pip install streamlit pandas numpy scikit-learn matplotlib seaborn xgboost
7. Save the app.py file to a local folder
8. In the terminal, navigate to the folder: cd path/to/folder
9. Run the app: streamlit run app.py
10. The app opens automatically at: http://localhost:8501

9. Technologies Used

Library / Tool	Version	Purpose in This Project
Python	3.x	Core programming language for all components
Streamlit	Latest	Build and serve the interactive web application
Pandas	Latest	Load CSV data, data manipulation and cleaning
NumPy	Latest	Numerical computations and array operations
Scikit-learn	Latest	StandardScaler, Linear Regression, Random Forest, metrics
XGBoost	Latest	XGBRegressor — extreme gradient boosting model
Matplotlib	Latest	Create all custom charts and figures
Seaborn	Latest	Scatter plots, histograms, correlation heatmaps
GitHub	-	Version control and source code hosting
Streamlit Cloud	-	Free cloud deployment and public URL hosting
Google Colab	-	Initial development and experimentation environment

10. Conclusion

This project demonstrates a complete, production-ready Data Science workflow — from raw data to a deployed web application accessible by anyone with a browser.

Starting from a dataset of 500 student records, we built an end-to-end machine learning pipeline that cleans and preprocesses the data, trains and evaluates three ML models, visualizes key data insights, and deploys the best model in an interactive bilingual web application.

Key Results

Achievement	Details
Best Model Accuracy	XGBoost with R^2 approximately 0.89
Most Important Feature	CGPA — strongest predictor of admission
App Languages	English and Arabic with real-time toggle
Models Available	3 models with plain-language descriptions for students
Deployment	Live at university-admission.streamlit.app

Future Improvements

- Expand the dataset with more universities and program-specific admission data
- Add support for international grading systems such as GPA out of 4.0
- Implement cross-validation and hyperparameter tuning for better model optimization
- Add a feature to compare multiple student profiles side by side
- Build a recommendation engine that suggests how to improve weak areas
- Add user login and history to track prediction changes over time

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